

Practical 1

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

int main()
{
    char command[100];
    pid_t pid;

    printf("Enter a Linux command: ");
    scanf("%s", command);

    pid = fork();

    if (pid < 0)
    {
        printf("Fork failed!\n");
        return 1;
    }

    if (pid == 0)
    {
        // Child Process
        printf("\nChild Process\n");
        printf("Child PID : %d\n", getpid());
        printf("Parent PID: %d\n", getppid());

        execlp(command, command, NULL);

        // Executes only if execlp() fails
        printf("Invalid command!\n");
        exit(1);
    }
    else
    {
        // Parent Process
        printf("\nParent Process\n");
        printf("Parent PID : %d\n", getpid());
        printf("Child PID : %d\n", pid);

        wait(NULL);

        printf("Child process completed.\n");
    }

    return 0;
}
```

Output-

```
ubuntu@ubuntu:~/OSLab/Practical1$ ./program_1
[Enter a Linux command: uname

Child Process
Child PID : 2491
Parent PID: 2490

Parent Process
Parent PID : 2490
Child PID : 2491
Linux
Child process completed.
```

100 # program_1

100 uname

101 lsblk

102 ps

103 top

104 history